A kernel-checked numerical certificate for a BerkeleyGW G_0W_0–GPP calculation

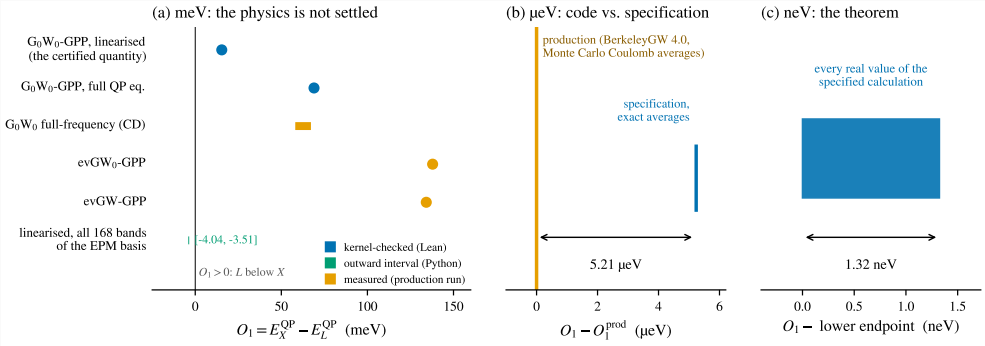
Si-EPM: what a proof assistant can and cannot say about a production GW number

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October 2026

Formalization, certificates and analysis code written by Claude (Anthropic) under the authors' direction; reviewed by a separate AI reviewer (Codex). Not a human-maintainer audit.

In one slide



- One production BerkeleyGW 4.0 run: `testsuite/Si-EPM`, GPP, 18 bands, 7 Ry.
- $O_1 = E_X^{\mathrm{QP}} - E_L^{\mathrm{QP}}$: 15 meV out of Σ's of ∼9 eV.

- Lean theorem, no hypotheses: the specified calculation on the run's own data lies in a 1.32 neV interval.
- **Production is within 5.21 μeV.**

What “verified” can mean for a GW number

Layer		Question	Here
1	implementation	does the program evaluate the specified finite calculation?	certified
2	convergence	is that finite calculation stable as controls grow?	measured only
3	model	is G_0W_0/GPP the right physics?	**not claimed**

A small layer-1 error says nothing about layers 2 and 3. We keep the three apart and never add them into one number. The headline is a *layer-1* statement about one input deck.

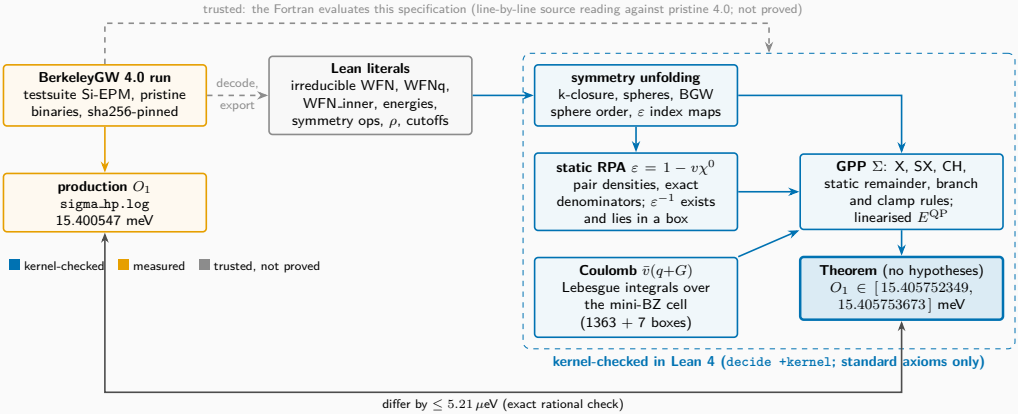
The test case: a deliberately fragile observable

- BerkeleyGW 4.0, `testsuite/Si-EPM`, shipped inputs, pristine binaries (archive sha256-pinned), real flavour, serial.
- EPM mean field, $V_{xc} = 0$: $E^{\mathrm{QP}} = E^{\mathrm{EPM}} + \Sigma$. Model silicon, not real silicon.
- O_1: conduction-band ordering X vs. L (each a multiplet average).
- In mean field X is 0.628 eV *below* L; G_0W_0 reverses it to $+15.4$ meV.
- O_1 is a 2–6 % residual of $X-L$ differences in exchange ($+0.71$ eV), SX−X (-0.43 eV) and CH ($+0.5$–0.6 eV).

Why this one

A quantity that looks stable while its parts are not. At the 1.5 Ry cutoff used in an early milestone, the same run gives $O_1 = -75.4$ meV: the sign flips with a single control.

The proof chain and its trust boundaries



The statement

```
theorem o1_enclosure_irr_unconditional :
    O1 glI stXI stLI EIrr vbTrue oqTrue ∈
      Icc ((8880811588953599 : ℝ) / 2 ^ 59) ((8880812351936001 : ℝ) / 2 ^ 59)
```

- `O1` is a real number defined by the Lean specification: χ^0 from pair densities with exact energy denominators, $\varepsilon = 1 - v\chi^0$ with $v = 8\pi/|q+G|^2$ and real π, the true ε^{-1}, GPP modes, X/SX/CH, the static remainder, the linearised QP equation.
- Inputs are the *irreducible* WFN/WFNq/WFN_inner coefficients; the unfolding to the full grid (k-closure, sphere order, ε index maps) is proved, not assumed.
- `vbTrue`, `oqTrue`: the exact mini-BZ averages $\frac{1}{|\Omega_q|}\int 8\pi/|q+G+p|^2\,dp$, Lebesgue integrals over the cell; $1363 + 7$ boxes proved.
- Checked by the Lean kernel with `decide +kernel`; the audit admits only `propext`, `Classical.choice`, `Quot.sound`. No `sorry`, no `native_decide`.

The specification reads like the Fortran

BerkeleyGW 4.0 `mtxel_cor.f90`, occupied GPP branch

```
wdiff = wxt - wtilde
delw  = wtilde * CONJG(cden) * rden ! = wtilde/wdiff
...
if (wdiffr.gt.limittwo .and. delwr.lt.limitone) then
  sch = delw * I_eps_array(ig,my_igp)
  ...
else if ( delwr .gt. TOL_Zero) then
  sch = 0.0d0   ! near pole
  ...
```

Lean `SigmaSpec.lean`

```
noncomputable def schM (I O x : ℝ) : ℝ :=
  if keepM I O then
    (let w := wtM I O
     if (x - w) ^ 2 > GAM2 ∧ w ^ 2 < (x - w) ^ 2 * LIM1
     then w * IeU I / (x - w) else 0)
  else 0
```

- Division-free tests, proved equivalent to the source's branch and clamp rule wherever the source's formula is defined (`certSsx_eq_srcSsx`).
- A generated guide (`docs/review/index.html`) puts every Lean definition next to the Fortran lines it transcribes. That correspondence is source reading plus numerical agreement, not a proof.

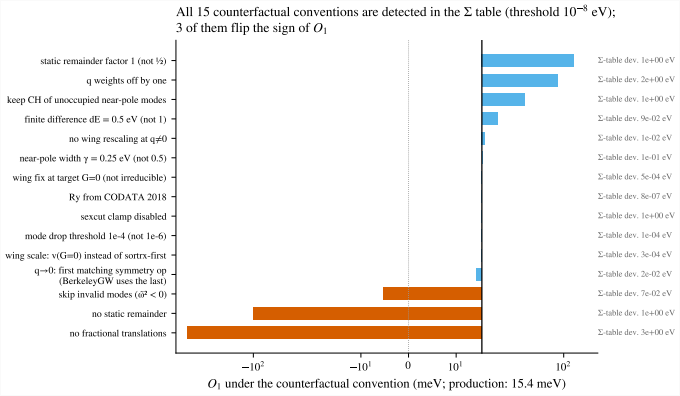
What the 5.21 μeV is

- Production evaluates $\bar{v}(q+G)$ by Monte Carlo (MT19937, seed 5000). The averages differ from the integrals by up to 1.7×10^{-3} (relative) and break star symmetry.
- Individual E^{QP} differ from the specification by ≈ 0.14–0.18 meV (CH by ≈ 0.14–0.30 meV). In O_1 these cancel to 5.21 μeV.
- With production's *own* Monte Carlo averages, the interval evaluator (Python, outward binary64) encloses the printed E^{QP} of all four states and O_1.
- The 5.21 μeV is an exact-rational comparison of the printed value with the kernel-checked endpoints (1 neV print-rounding allowance).

Reading

On this deck, BerkeleyGW 4.0 computes exactly what its source says, to within its Monte Carlo quadrature. No implementation discrepancy was found on any active code path.

Which conventions decide the answer



- Each bar: re-evaluate with one convention changed. Every change is *detected* in the Σ table at 10^{-8} eV (the specification is that tight).
- These are **not** bugs. They show which documented choices carry the sign of O_1: the static remainder (-101 meV without it), invalid-mode handling (-5.2 meV).
- Usable as a regression oracle for convention changes.

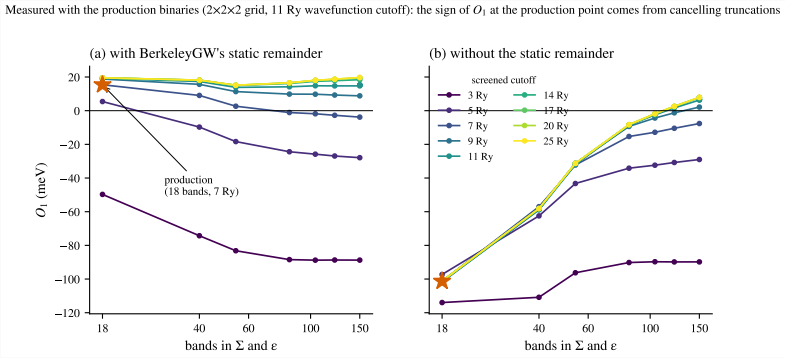
The $q \to 0$ cell: one direction of an anisotropic ε^{-1}

- Σ carries ε^{-1} to the full q-grid by the space-group formula and, for each q, uses the *last* matching (iq, op) pair. For $q = 0$ that is op 48 of 48.
- The head and wings at $q_0 = (0, 0, 0.001)$ are direction-dependent, so this choice selects one direction:
  - over all directions BerkeleyGW could have picked, $O_1 \in [14.34, 15.41]$ meV;
  - the direction average (cubic crystal) gives 14.89 meV;
  - exactly degenerate states split: the Γ valence SX$-$X spreads by 36 meV before the multiplet average hides it.
- Other facts that surfaced: the ε k-sum runs over the 32-point symmetry closure of the shifted $2\times2\times2$ grid, not the nominal 8; wings at $q \neq 0$ are rescaled through the *irreducible* $G = 0$ and the sortrx-first v.

These numbers are outward-interval (Python) results, not kernel-checked.

Layer 2: the sign comes from cancelling truncations

Measured with the production binaries (2×2×2 grid, 11 Ry wavefunction cutoff): the sign of O_1 at the production point comes from cancelling truncations



- At 7 Ry, adding bands drives O_1 from $+15.4$ to -3.8 meV; raising the cutoff at 18 bands looks converged. The static remainder carries the sign. 125 production runs. Measured; no convergence claim is made.

Layer 3: how far the GW hierarchy moves O_1

treatment	O_1 (meV)
G_0W_0-GPP, linearised	$+15.4$
G_0W_0-GPP, full QP equation	$+69.0$
G_0W_0 full frequency (CD)	$+58$ to $+67$
evGW_0-GPP	$+137.9$
evGW-GPP	$+134.2$
SOX (leading vertex), estimate	$+2.0$ shift

- $V_{xc} = 0$ puts E^{QP} $\sim$7 eV from E^{EPM}, deep in Σ's pole structure: one Newton step moves O_1 by -53.6 meV.
- Proved: in $[-10^4, 10^4]$ eV each state's full GPP QP equation has *exactly two* roots ($Z = 0.86$ and 0.04). Uniqueness of the principal root depends on the `sexcut` clamp; without it, 11 sign changes for X.
- Plasmon-pole error on $\Sigma_c(X) - \Sigma_c(L)$ at $E = E_o$: -31.6 meV vs. CD.

From theory to the specification: conditional identities

- Proved in Lean, under named hypotheses: the pole signs of Σ from the damped Fourier transform of time-ordered G_0; the HL-GPP kernel algebra and SX/CH split; the static χ^0/RPA structure (the factor $4 =$ spin $\times$ two time orderings).
- Every place where the code departs from the theory is named in a departure table: wings at $q = 0$, Coulomb averaging, mode-drop thresholds, ..., each "proved equal" or "named approximation".
- What formalization exposed:
  - at $q = 0$, $i\omega = 0$ the argument x is exactly 0 for the state itself; the code substitutes `TOL_Zero`, so the identity covers $\Sigma(E_o + dE)$ but not $\Sigma(E_o)$;
  - the GPP kernel is discontinuous at $(x, \Omega^2) = (0, 0)$: modes dropped because rounding makes $|\Omega^2|$ tiny are exactly right only in the ideal metric;
  - the one-system theorem cannot be instantiated at X or L (time-reversal-invariant points; a formal obstruction, `hyps_unsat_of_selfpaired`).

These identities are *not* instantiated on the certified run. They are not a derivation of the GPP f-sum prescription.

What is not claimed

Trusted, not proved

- decoding and export of the run files into Lean literals;
- that the Fortran evaluates the specification (source reading on every active path);
- the production execution itself.

Not claimed

- that BerkeleyGW is correct on any other input;
- that 18 bands are converged;
- that GW/GPP or the EPM describe silicon;
- formal verification of BerkeleyGW's physics.

Trusted computing base: the Lean 4.19 kernel (with GMP `Nat`), Mathlib, the elan installer. No external re-checker (lean4checker) has been run yet.

Why this might be useful to BerkeleyGW

Now

- An executable, line-referenced specification of the real-flavour GPP Σ path.
- A reference value with a proof-backed enclosure for a testsuite deck.
- A convention oracle: 15 counterfactuals, each detected at 10^{-8} eV.

Next, with your input

- Does the specification match your reading of the code? (Most valuable feedback.)
- Which target next: LDA V_{xc}, complex flavour, full frequency, a 2D material with truncation?
- Would certified reference values be useful in the testsuite?

Scale and reproduction

theorems in the axiom audit	34 553
certificate-data modules	1 376 (63 MB)
Lean outside data	∼37 k lines
Coulomb boxes proved	1 363 + 7
production runs (layer 2)	125
full gate, cold build	∼5 h

```
bash scripts/bootstrap.sh
bash scripts/verify.sh
```

`bootstrap.sh`: user-space Lean 4.19 and Mathlib.
`verify.sh`: build, exhaustive axiom audit, 699 scientific and 16 workflow tests.

Reviewer's guide: `docs/review/index.html`.
Every caveat: `docs/DETAILS.md`, `docs/ENDPOINT.md`, `docs/BRIDGE.md`.

A production GW number, checked to 5 μeV against a proved enclosure.

Everything else (convergence, the plasmon pole, the vertex) remains physics.

References

- M. S. Hybertsen and S. G. Louie, Phys. Rev. B **34**, 5390 (1986). GPP model.
- J. Deslippe *et al.*, Comput. Phys. Commun. **183**, 1269 (2012). BerkeleyGW.
- B.-C. Shih *et al.*, Phys. Rev. Lett. **105**, 146401 (2010). Coupled convergence in bands and cutoff.
- The Lean 4 theorem prover and Mathlib.

Backup: evidence for each number in this talk

number	evidence	where
$O_1 \in [15.405752349, 15.405753673]$ meV	Lean, no hypotheses	`Headline.o1_enclosure_irr_unconditional`
$O_1 > 0$	Lean	`Headline.o1_pos_irr_unconditional`
full-QP $O_1 \in [68.9856742, 68.9856744]$ meV	Lean	`Headline.o1_fullQP_irr_unconditional`
5.21 μeV	exact rationals vs. Lean endpoints	`tests/test_endpoint_e1.py`
$q \to 0$ range, spec B, CBM at L	outward interval	`bgw/endpoint/cert.py`
all-band $O_1 \in [-4.05, -3.51]$ meV	outward interval	`bgw/r28_f3`
convention study	reproducer vs. production	`data/mutations_e1.log`
convergence, FF, evGW, SOX	production runs	`docs/ENDPOINT.md`